DynoSure

AUTOMOTIVE CAN BUS SOLUTIONS

Comprehensive Product Catalog & Specifications Guide

Affordable, reliable diagnostic and data logging hardware engineered specifically for the Indian automotive sector. Based on open-source standards with vendor lock-in free integrations.

DynoSure India
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DynoSure SLCANv1

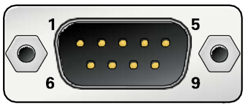
Model: SLCANv1 (USB-to-CAN Adapter)

The DynoSure SLCANv1 adapter provides a reliable and convenient connection between a PC and a CAN bus. Based on the open-source CANable2 firmware and the Lawicel SLCAN protocol, it exposes the CAN interface as a standard virtual COM port, simplifying integration with existing tools and reducing software overhead.

Specifications

Parameter	Details
Microcontroller	STM32G4 Series, 170 MHz
CAN Protocols	CAN 2.0A (11-bit), CAN 2.0B (29-bit), CAN-FD
USB Interface	USB 2.0 Full-Speed (compatible with USB 1.1 / 3.0)
Standard Bitrates	5 kbps to 1 Mbps
CAN-FD Bitrates	Up to 8 Mbps data phase bitrates
Power Supply	USB-powered (no external supply required)
Compatibility	Windows & Linux (exposes standard Virtual COM Port)

DB9 Pinout Assignment



DB9 Pin	Assignment
Pin 2	**CAN-L** (CAN Low)
Pin 3	**GND** (Ground)
Pin 6	**GND** (Ground, secondary)
Pin 7	**CAN-H** (CAN High)
Others	Not Connected

DynoSure SLCAN GPIO

Model: SLCAN GPIO (USB-to-CAN with 8-ch GPIO Control)

The DynoSure SLCAN GPIO combines a full-featured USB-to-CAN adapter with 8 configurable GPIO outputs. This gives hardware developers and test engineers both CAN bus communication and physical hardware control in a single compact device. Exposes CAN standard interfaces and digital outputs directly through internal messaging controls.

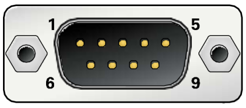
Specifications

Parameter	Details
Microcontroller	STM32G4 Series, 170 MHz
CAN Protocols	CAN 2.0A (11-bit), CAN 2.0B (29-bit), CAN-FD
USB Interface	USB 2.0 Full-Speed
GPIO Channels	8 Configurable Digital Outputs
Power Supply	USB-powered (no external supply required)

GPIO Control Protocol

Field	Value / Description
CAN ID	**0x1FFFFF** (Extended 29-bit ID)
DLC	**2**
Byte 0	GPIO States (Bit 0 = GPIO 0 ... Bit 7 = GPIO 7). 1 = HIGH, 0 = LOW
Byte 1	**0xAA** (Fixed GPIO command identifier)

DB9 CAN Pinout



DB9 Pin	Assignment
Pin 2	**CAN-L** (CAN Low)
Pin 3	**GND** (Ground)
Pin 7	**CAN-H** (CAN High)
Others	Not Connected

DynoSure LoggerV1

Model: LoggerV1 (Standalone CAN Data Logger)

The DynoSure LoggerV1 is a standalone CAN bus data logger designed specifically to capture CAN 2.0 traffic to onboard storage without requiring a connected PC during operations. It logs data in industry-standard Vector ASC and MDF4 (.mf4) file formats, avoiding restrictive closed ecosystems and vendor lock-in.

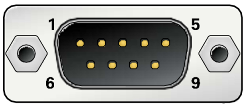
Specifications

Parameter	Details
Microcontroller	Raspberry Pi RP2350
CAN Protocols	CAN 2.0A (11-bit ID), CAN 2.0B (29-bit extended ID)
Storage	Onboard microSD card (FAT32 filesystem)
Log Format	Vector ASC (.asc) & MDF4 (.mf4)
USB Interface	USB 2.0 (Mass Storage Device / Card Reader mode)
Power Modes	• **Logging Mode**: Requires **+12V DC** external supply • **USB Mode**: USB-powered (acts as SD card reader)
Firmware Update	RP2350 drag-and-drop .uf2 file

microSD Configuration (configuration.txt)

Key	Options / Values
Bitrate	**1** = 500 kbps (default), **2** = 1 Mbps, **3** = 250 kbps
FileFormat	**0** = Vector ASC (.asc), **1** = MDF4 (.mf4)
ProgramMode	**0** = Logging, **1** = Program Mode (RP2350 .uf2)

DB9 CAN & Power Pinout



DB9 Pin	Assignment
Pin 2	**CAN-L** (CAN Low)
Pin 3	**GND** (Power Ground)
Pin 7	**CAN-H** (CAN High)
Pin 9	**+12V DC** (Power Input)
Others	Not Connected

DynoSure OBD to DB9 Converter

Model: OBD2DB9 (Universal Diagnostic Connection Cable)

The DynoSure OBD to DB9 Converter provides a universal and reliable connection between your vehicle's OBD-II port and standard CAN analysis tools. It routes standard CAN bus lines and vehicle battery power directly to the correct pins on your device, allowing you to connect and power your SLCANv1, SLCAN GPIO, or LoggerV1 interfaces directly from the vehicle's diagnostic port without additional wiring.

Specifications

Parameter	Details
Cable Length	1.0 Meter (Heavy-duty molded strain relief)
Conductors	Pure low-resistance copper for high signal integrity
Compatibility	DynoSure SLCANv1, SLCAN GPIO, LoggerV1, and standard tools
Power Delivery	Routes vehicle battery power (+12V) directly to DB9 Pin 9

Pinout Mapping Table

DB9 Pin	OBD Pin	Signal	Description
Pin 2	Pin 14	**CAN-L**	CAN Low line
Pin 3	Pin 4 & 5	**GND**	Chassis/Signal Ground
Pin 7	Pin 6	**CAN-H**	CAN High line
Pin 9	Pin 16	**+12V**	Battery Power Out

DynoSure CANviz

Product: Browser-Based CAN Bus Analyzer

DynoSure CANviz is a modern, open-source CAN bus analyzer. Based on canviz by Chanchaldhiman, it runs entirely in your local browser and installs with a single pip command. Works with any standard USB-to-CAN hardware adapter, including SLCANv1 and SLCAN GPIO, putting frames, signals, and protocol decoders in front of you instantly.

Features

• **Live Message Table**: virtual scrolling handles up to 2,000 frames/sec with zero drops.
• **DBC Decoding & Plotting**: Upload a DBC file to plot signals in a live time-series graph.
• **Passive Decoders**: Passive J1939 and CANopen (CiA 301 + 402) decoders built right into the user interface.

Getting Started

pip install dynosure-canviz
canviz

Information

Attribute	Details
Platform	Windows, Linux, macOS
PyPI Package	dynosure-canviz
Source Code	GitHub Repository
License	MIT Open Source